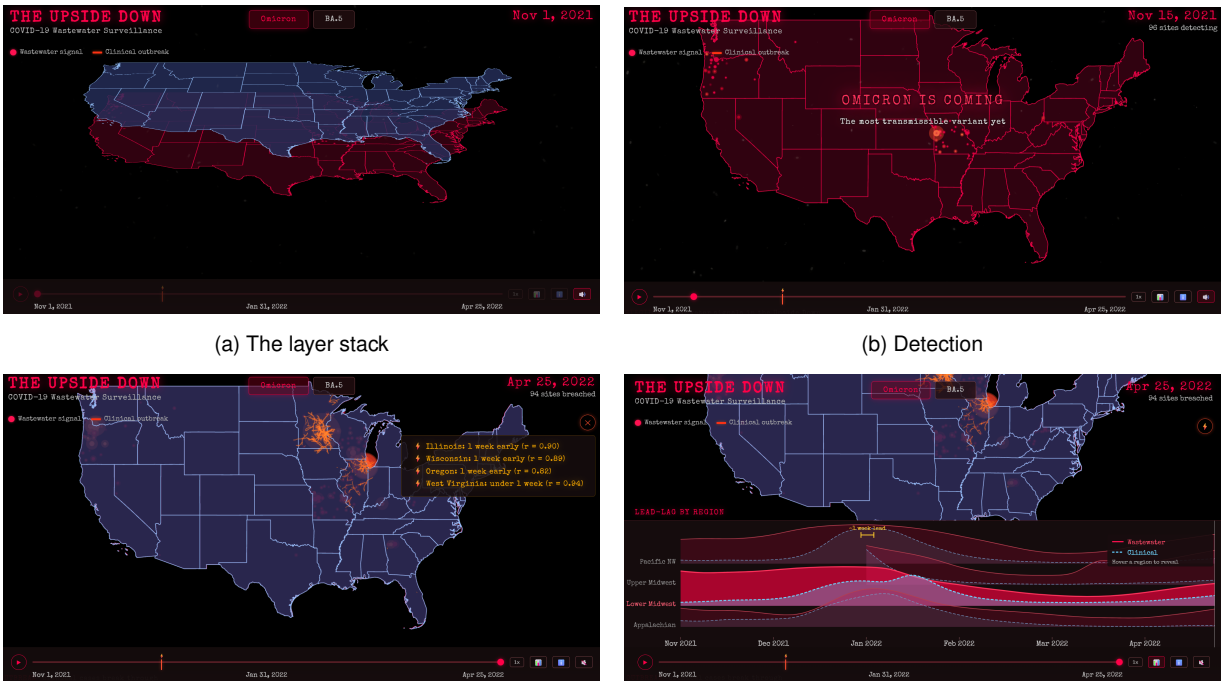
The Lead-Lag Layer Stack: Encoding Temporal Precedence as Spatial Depth in *The Upside Down*

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(a) The layer stack

(b) Detection

(c) The breach

(d) Lead-lag ridgeline

Figure 1: The piece in four moments. The scene tilts and the two map layers pull apart in depth, the lower red map carrying wastewater and the upper blue map clinical cases. Sampling sites then light up below while the surface above stays intact. At the breach the surface returns and cracks spread across it, concentrated where wastewater detected most. The ridgeline opens underneath and marks the measured lead for one region. The palette, typography and controls carry the look of the show throughout.

Abstract

Wastewater surveillance can detect SARS-CoV-2 outbreaks days to weeks before clinical case counts rise, yet public dashboards show this through static line charts and choropleth maps. The general problem is that when one signal in a place arrives before another, a visualization has to convey both where the signals are and which came first. We present *The Upside Down*, an animated data storytelling experience built around two encodings. The first draws the signals as transparent map layers over the same geography that pull apart in depth, putting the earlier signal underneath the later one, which we call a *lead-lag layer stack*. The second sets both curves in each row of a ridgeline plot and marks the gap between them, so the lead is stated rather than judged by eye. Drawing inspiration from the *Stranger Things* series, *The Upside Down* turns on a hidden world lying beneath the visible one, where danger appears before it surfaces. That is the same shape as the data relationship, and it is what the layer stack is built from. Rather than emulate a science fiction interface, this project incorporates the central premise of a sci-fi show, along with its production design (e.g., palette, typography) that carries its horror.

Index Terms: Lead-lag encoding, spatiotemporal visualization, figurative frames, science fiction, wastewater surveillance.

1 Introduction

This workshop asks how science fiction and visualization inform one another. Often that exchange runs through interfaces. Film and television invent interfaces that have never been built, and Shedroff and Noessel traced those fictional designs into products people have actually seen and used [14]. We take a different route. A story's premise can also inform visualization and interface design.

In this project, we consider wastewater surveillance data. Infected individuals shed SARS-CoV-2 RNA into wastewater before symptoms appear and before they seek testing [2, 4], so outbreaks can be detected days to weeks earlier than hospital reporting allows [3]. Public-facing visualization of this data often incorporate time-series overlays and choropleth maps. Those forms serve analysts well. Our own reading is that they do not serve the general public, because what makes this data worth seeing is not a case count but an order of events. The virus was detectable underground

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well before it showed up in hospitals, and neither a line chart nor a choropleth is built to show this.

The problem is not specific to COVID-19. Wherever one spatial signal reliably precedes another across the same geography, a visualization must convey both where the signals are and which arrived first. We contribute two encodings of this phenomena, both inspired by the Netflix series *Stranger Things*, along with a storytelling interface that carries them. Its typography, colour palette, and interactive controls follow the production design of the show, which is set in the 1980s, so the piece reads as an interface of that decade rather than as a modern dashboard.

In *Stranger Things*, the *Upside Down* is a second version of the same town, occupying the same streets and buildings but is dark and decaying. It stays out of sight until a crack opens between them, and whatever is inside the Upside Down has been spreading long before that happens. Two aspects of the Upside Down inspire our design. The two worlds share a geography, and one of them temporally precedes the other.

Our **lead-lag layer stack** design builds on both aspects. It draws the two signals as transparent planes over the same geography. When laid flat they coincide, and the display reads as an ordinary map. When tilted they separate in depth, and the gap between them is the relationship (Fig. 1a). The viewer sees which signal came first by looking at which plane sits underneath, instead of comparing two charts side by side and working out the order.

A **paired lead-lag ridgeline** gives each geographic region a row, with a filled curve for the signal that arrives first and a dashed curve for the one that follows. With one row per wave, the gap between the two peaks is marked directly (Fig. 1d), so the lead time is something the reader is directed to rather than left to locate on their own. This plot has no visual counterpart in the show, but it reaches the same idea by another route. Stacked traces like these look like a seismograph, and a seismograph reads the ground before it breaks open. That is the same moment the cracks draw, when the Upside Down breaks through into the ordinary world.

2 Related Work

Tilt is not new to maps. Yang et al. [5] use it to swap between a choropleth map, a prism map and a bar chart, where the act of tilting changes the visual encoding. In contrast, our use tilt exposes a temporal gap indicating the outbreak lead time.

Tilt also has precedence in epidemiology. Te-Chaniyom [7] tilts a map to fold time into a two dimensional geography for disease clusters, and reports that the tilted view reads more easily than the flat one. That work tilts to make room for a time axis. We tilt to separate two signals that already share one. Meanwhile, ridgeline plots are common when analysts compare epidemic curves, but they rarely appear in material written for a general audience. However, neither tilted maps or ridgeline plots appear in conventional public health communication.

The lead-lag relationship in epidemiology bears similarity to one in Earth science. Seismic swarms, ground movement, and gas emissions all arrive before an eruption, an interval that can be measured and shown [6]. This practice os measurement reads the surface to infer what is below. Wastewater analysis inverts this, reading what is below to predict the surface, which is why two stacked layers are well-suited for this analysis.

3 What the Frame Encodes

The *Stranger Things* inspiration operates at two levels, and separating them matters.

The first is structural. In the *Stranger Things* fictional universe, a second world is superimposed in the same coordinates as the ordinary one, and this is detectable before it breaks through. Nothing in that depicts an object, a person or a place. It is a conceptual metaphor, a familiar arrangement borrowed to hold something abstract.

Visualization has long incorporated metaphor, ranging from the abstract or conceptual to the specific. Hierarchies have been drawn as growing plants [9], scalar fields as terrain [11], and neural connectivity as a subway map [8]. These choices are not neutral. Woodin et al. [12] show that conceptual metaphor shapes how readers interpret even an ordinary line graph, and Pokojná et al. [10] trace the same machinery through scientific storytelling.

In those systems the metaphor is a layout. It gives the data a shape the viewer can move through, but the data would still hold if it were drawn another way. Ours is not a layout. The distance between the two planes is the lead time, so when the lead disappears the metaphor has nothing left to show.

The second level is figurative. A crack depicts something, a surface giving way, and it appears at the moment the two worlds meet. Its severity combines a state's case intensity with the wastewater signal at that site, so the places that detected most crack hardest. Byrne et al. [1] argue that figurative elements extend a visualization's expressive range beyond decoration. The cracks do both at once. They tell you how badly a place was hit, and they make that legible as a rupture rather than as a value.

The dots on the map (Fig. 2) carry two readings at once. Size and brightness follow the viral signal, so a light strengthens as the threat grows, which is how the show uses it. Blinking is set separately by coverage. Idaho contributes one wastewater treatment plant serving under three percent of the state against Illinois with seventy-one covering sixty-two. So Idaho's detection sites blinks on and off versus Illinois' sites pulsing steadily.

4 Animation Design

Each wave plays as five acts, following Segel and Heer's martini glass structure [13], which opens with an authored sequence before releasing the viewer to explore.

Entry. The scene tilts to 65 degrees and the two maps separate in depth. The surface layer drifts up and away while dust fills the void between them. The scene flattens, leaving the red layer alone. This eleven second sequence establishes the two worlds before any data arrives.

Detection. Wastewater dots grow and brighten across the map (Fig. 1b). Partway through, the camera flies to a thinly sampled state, Arizona during Omicron and Montana during BA.5, where the blinking is the only thing on screen. This is where coverage is taught.

The Breach. Playback pauses, the scene tilts again, and the blue surface returns and drops back into place (Fig. 1c). Only then are cracks enabled, so the viewer watches them grow from nothing.

Zoom. The camera flies to the wave's focal state, Illinois for Omicron and Georgia for BA.5, while the ridgeline slides up from below. Lead times appear before the camera pulls back to the full map.

Exploration. The lead time panel stays open. The viewer can replay, switch waves, scrub freely, or reveal the measured lead for every state in the wave.

5 How the Data Shaped the Design

Only two variants are profiled in the interface and a viewer may want to know why. What the data could support decided what the design could show. The Delta wave had to be removed because CDC NWSS percentile reporting does not begin until 15 November 2021, so the entire Delta window contains no viral intensity data. Within the two waves that remain, not every region behaves alike. Some show a clear gap between the curves, some show none, and in one case the clinical peak arrives before the wastewater peak.

Our strongest numbers came from our weakest data. Ranked on correlation alone, Idaho led Omicron by three weeks at $r = 0.99$,

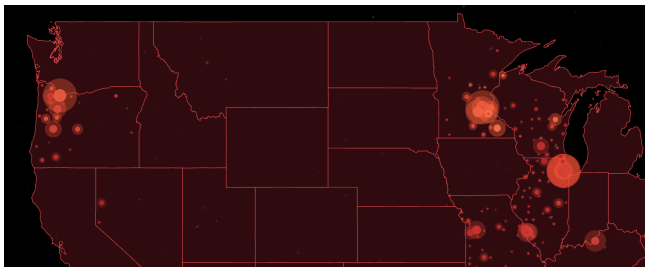


Figure 2: Detection phase. Wastewater sites appear as glowing dots before any surface cracks. Sites in thinly sampled states render unstable rather than confident.

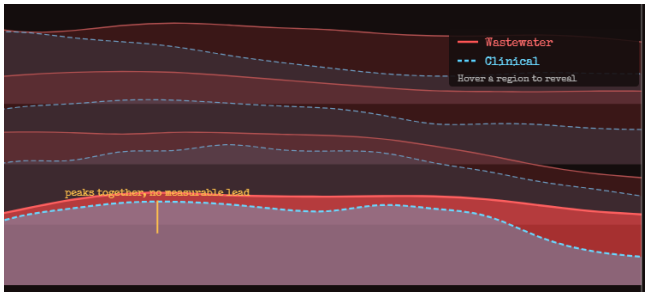


Figure 3: The counterexample. With Appalachian highlighted the two curves are coincident. No gap, no lead time, nothing for the layers to separate.

on one wastewater management plant. We now report a state only when its sampled plants cover at least 15% of the population across five or more sites. Every large lead time disappeared under that rule. What survives is modest. Where a lead is measurable it spans about one week. During BA.5 most adequately covered states fell below that, matching the range reported in the literature [2, 4]. This is a tradeoff regarding attention. Three weeks is a more dramatic finding than one week. Yet applying the coverage rule left us with a version that more closely matched the reported lead-lag range. The animation directs viewer attention instead, by controlling the order things appear in rather than by the size of any number.

The encoding also reflects a floor. Our wastewater readings are daily, but the case counts are weekly, and a lag can never be finer than the coarser series. Published leads run from roughly two to eight days [2, 4], which sits at or below that limit. So a row marked under one week is not a row without a lead. It is a row whose lead is too short for this data to measure. We could have drawn a small gap anyway and few viewers would have questioned it. Drawing nothing is the more honest choice, despite being less satisfying as a storytelling experience.

Appalachia, which in this data is effectively West Virginia, remains as an apparent counterexample (Fig. 3). Its two curves sit on top of each other and the layers have nothing to separate. This unevenness is not a flaw in the visualization but a reflection of the real variability documented by Kumar et al. [3].

We could have shown only the regions with the clearest gaps between the two curves. The piece would have looked more convincing and told the viewer less about how wastewater surveillance actually behaves. Keeping Appalachia means the depth encoding is left with nothing to draw. There is no lead, so there is no separation between the layers, and the metaphor returns an empty result instead of a persuasive one. That is the evidence that it was carrying the data rather than sitting on top of it.

That capacity to return nothing is what we would want from any conceptual metaphor used this way. Its credibility rests on the cases where it does not hold. That matters more here than it would in an analytical tool. An analyst reads the numbers and treats the metaphor as a convenience. A public audience reads the metaphor and takes the numbers on trust, which is why the metaphor has to answer to the data. A metaphor that cannot fail against real data is closer to concept art, which is valuable but a different contribution.

6 Limitations

The clearest limitation is a design choice rather than a data problem. Narrative framing introduces a tradeoff between engagement and precision. By prioritizing storytelling and emotional resonance, the piece gives up some of the analytical exactness a standard CDC dashboard provides. A viewer who leaves understanding that wastewater can lead clinical cases by about a week has gained the core insight, but they cannot read exact case counts or compare percentile values across states. This is intentional for the audience we are designing for, and it means the piece complements expert-facing tools rather than replacing them.

A second limitation follows from the source of our inspiration. *Stranger Things* straddles the genres of science fiction and horror, and the latter is not a desirable register for public health communication. The intended outcome is not that viewers feel alarmed but that they leave feeling informed and able to use publicly available wastewater data for their own communities. Whether the framing delivers that or simply frightens people is something we have not yet examined.

Remaining limits are technical or related to dataset availability: the stack needs WebGL and hardware-accelerated transforms, we report correlations rather than causal estimates, sewer transit time and sampling frequency are not modeled [3], and coverage spans Omicron and BA.5 only.

7 Discussion and Conclusion

Stranger Things is unusually well-suited to this kind of work. It blends science fiction, horror, and fantasy, so a viewer arrives with several types of expectation and none of them are about data. Its production design is specific enough to be recognized from a colour palette and a typeface alone. The show is set in the 1980s, so borrowing its design conjures the GUI interfaces of that decade (c.f., GUIs shown in the film *WarGames*, released in 1983). Our interface is therefore retro-futurist rather than futuristic. Many interfaces inspired by science fiction look to work set in the future. In contrast, we plot contemporary data in an interface evoking another period.

A conceptual metaphor is not merely an aesthetic choice but a structural decision that shapes how viewers interpret data relationships. *The Upside Down* does not just decorate the lead-lag relationship. It makes that relationship spatial, placing the virus below in wastewater before it appears above in clinical cases. The closer a metaphor's structure sits to the shape of the data, the more work it can do.

We presented *The Upside Down*, an animated data storytelling experience that uses a conceptual metaphor to communicate the early warning potential of COVID-19 wastewater surveillance. Wastewater and clinical data are rendered on two transparent map layers that separate in depth, synchronized with a ridgeline plot that marks regional lead times. Effective public health visualization should not just display data but empower people to act on it. Before that can happen, audiences need to feel confident that they understand what they are looking at. By grounding the piece in a story viewers already recognize, The Upside Down builds that confidence first, so that when someone meets wastewater surveillance data in the real world, they have the context to pay attention to it.

The application and its source are available at `https://the-upside-down.netlify.app/`.

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